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RESEARCH ARTICLE

Developing a Robust Question and Answer System for the Turkish Language Utilizing Deep Learning Techniques

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ABSTRACT Research on open-ended question and answer systems faces several complex challenges within the domain of natural language processing, one of which is limited data availability. Although numerous studies have been conducted in widely spoken languages such as English, there is a notable scarcity of research in languages such as Turkish. In our preliminary investigation, we proposed a solution to address the data shortage issue in Turkish by translating the English SQuAD dataset into Turkish. Another challenge is the success of deep learning models that use large language models. We developed various baseline models using deep learning techniques on this newly created dataset and performed analyses from multiple perspectives. Deep learning models and large language models often present an architectural enigma for many researchers, so analyzing both the questions and the corresponding answer-bearing data is of utmost importance. We have shown that the structuralization and enrichment of the data contribute significantly to the success of the model. In our research, we devised an architecture that incorporates a structural transformation of data before use in model training. This approach enabled us to enhance the learning capacity of the system without altering the underlying closed-box architecture of the large language models and deep learning systems employed.

INDEX TERMS Natural language processing, question answering, machine reading comprehension, Turkish question answer dataset, SQuAD, knowledge base data, large language models, deep learning.

I. INTRODUCTION

The domain of question-and-answer systems presents a complex challenge within natural language processing and has gained considerable interest in recent years. Although extensive research has focused on widely spoken languages such as English, studies in languages such as Turkish are still limited. This research seeks to provide a thorough, practical, and innovative road map for creating question-and-answer systems in Turkish, aiding in the progress of such systems for underrepresented languages.

A significant obstacle hindering academic research in this field is the scarcity of accessible datasets in Turkish. Developing a comprehensive dataset for open-ended question-and-answer systems is a labor-intensive process that usually requires extensive collaboration between large teams. To address this issue, our preliminary study focused on translating the widely recognized SQuAD data set into Turkish using machine translation methods. After thorough manual revisions, we evaluated the consistency and quality of the newly created dataset using baseline deep learning models. Detailed results related to this dataset can be found in the publication accessible via the UBMK 2024 Conference[1] and [1].

The research carried out served as a valuable reference for our analysis. We evaluated the consistency of the dataset with

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[1]UBMK, 9th International Conference on Computer Science and Engineering, https://ubmk.org.tr/en/

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previously translated datasets. Through the implementation of both manual and batch processing techniques, we successfully utilized 78% of the original data. We evaluated the model's performance by constructing deep learning models based on this dataset and identified key areas requiring further attention.

Our analysis revealed that the performance of the model varies significantly between different domains within the dataset. We found that while the model achieves exceptionally high accuracy in certain areas, its performance diminishes in scientifically oriented domains that are inherently complex and laden with technical terminology. Furthermore, we note that the complexity and type of questions posed also influence the model's ability to identify the corresponding answers accurately. Additionally, another critical observation is that accuracy tends to decline as both the length of the question and the corresponding paragraph increase.

These factors, such as domain, content, and type of questions, as well as the length of inputs, significantly influence the efficacy of the model. Although the same model can achieve considerable success across varying parameters, it may yield suboptimal results when there are changes in the domain, modifications to the questions, or an increase in content length.

The primary components of deep learning systems are large language models (LLMs) and deep learning architectures. However, the intricate nature of these systems presents significant challenges for intervention. For numerous developers, reconstructing an LLM or designing transformer-based deep learning frameworks with distinct functionalities requires highly specialized and complex efforts.

To create a more balanced and healthier model capable of adapting to various factors, we focus on the structural transformation of the data (questions & paragraphs). Using structurally transformed data for model training, we found that the models' learning capabilities improved, and these findings were thoroughly analyzed.

II. RELATED WORKS

A. DATASET

There are different types of question and answer tasks, one of which is Machine Reading Comprehension (MRC). MRC tasks are designed to extract answers from a given paragraph. SQuAD, released by [2], serves as a complete open-domain MRC dataset in the English language. Creating a comparable resource for a different language requires considerable effort. SQuAD v1.1 is an extensive machine reading comprehension dataset, featuring more than 100,000 questions that are crowd-sourced from Wikipedia entries. The training dataset comprises 442 titles, while the test dataset comprises 48. Each title includes numerous paragraphs and each paragraph contains multiple questions, some of which can have more than one answer. In our previous study [1],[2] the SQuAD dataset was translated into Turkish using automated machine translation. Subsequently, 78% of the original English dataset was successfully created in Turkish by applying editing and batch operations.

Gemirter and Goularas [3] previously translated the SQuAD into Turkish with 50% data availability. To assess the effectiveness of our translation method, the Gemirter dataset from [3] served as a control dataset in our previous study [1]. A different study [7] used Amazon Translator to translate SQuAD2.0 into Turkish. They translated the titles, context paragraphs, questions and answer spans in the original dataset, resulting in 25,528 question-answer pairs, along with 259 paragraphs without any question–answer pairs, as well as 3,582 question–answer pairs from the development data set. In total, they created approximately 50% usable original dataset. They made significant contributions to this field by using the dataset obtained from the question and answer system, which is based on an information retrieval architecture.

The THQuAD [4] dataset has been developed in recent years, specifically for the Turkish language. It follows the same format as the SQuAD dataset; however, it currently lacks the depth of content found in SQuAD. However, it represents a significant effort to establish a new open-domain question-answering dataset for Turkish. The dataset contains 15,554 questions, but is limited to only two domains: Ottoman History and Turkish Islamic Science History. The richness of content plays a crucial role in question-answering systems, as models may perform well on general subjects (such as countries, cities, and biographies of notable individuals), but may not exhibit the same level of success with more specialized topics, including philosophy and science.

Similar studies have been conducted in various languages. Reference [5] discussed the creation of SQuAD translated into Spanish. For this, a custom infrastructure was used for the translation process. Following translation, specific heuristics were applied to clean and enhance the retrieved (context, question, answer) examples, resulting in two datasets: SQuAD-es v1.1, which is larger but contains more noise, and SQuADes-small, comprises 53% of the original data and being smaller but still more reliable. Another study [6] was related to French. This research involved translating a subset of SQuAD into French and extracting multiple paragraphs along with their respective question-answer pairs, yielding a total of 327 pairs (question, answer, paragraph) in 48 articles, indicating that the findings are quite limited.

B. QUESTION ANSWER DEEP LEARNING MODEL STUDIES

Neural network architectures have made substantial advancements in the effectiveness of question-answering tasks owing to their linguistic representations and robust attention mechanisms. Reference [8] introduced a mechanism that suggests that some inputs should be prioritized over others, and the model supports this emphasis. In 2018, Google released Bidirectional Encoder Representations of

[2] Our dataset github link, https://github.com/kadirtutar/datasetTranslated

Transformers (BERT) [9]. This architecture is one of the most significant improvements in the field for achieving a remarkable accuracy rate on the SQuAD dataset. According to the current state-of-the-art table for SQuAD,[3] BERT achieved an exact match (EM): 0.87 and F1 score: 0.93 in English.

Furthermore, various neural network methods utilizing pre-trained language representations exist, including ELMo [10] and Generative Pre-trained Transformer (GPT) [11]. ELMo features a bidirectional architecture; however, adapting it to different tasks is more challenging than other solutions. GPT required fewer architectural differences and was unidirectional. Chat-GPT is one of the most important developments in this field [12]. It is a chatbot powered by generative artificial intelligence, created by OpenAI and introduced in 2022. It operates different large language models. ChatGPT can produce conversational responses that mimic human interaction, allowing users to adjust and guide the conversation according to their preferred length, format, style, detail level, and language.

Reference [3] is an important question-answering study in Turkish. They fine-tuned the BERT model on different question-answering datasets, also the study has used different language models BERTurk[4] and custom LLMs. BERTurk is one of the research projects conducted to develop a large languge model for the Turkish based on BERT. In our research, BERTurk served as a large language model. They achieved EM: 0.55 and F1 score: 0.67 on a limited SQuAD dataset. Another recent study [4] has developed models in the THQuAD dataset. They fine-tuned some neural network models and their results are Electra EM: 0.63 F1 score: 0.81, BERT EM: 0.62 F1 score: 0.81, Albert EM: 0.48 F1 score: 0.69. In addition, a rule-based system for question analysis in Turkish [13] concentrates on dissecting the question components of the question-and-answer system. After evaluating a question, it generates focused information derived from that question. The identified focal information can then be used to search for answers across various data sources. Generalizing rule-based studies and feature extraction methods for open domains demands considerable human effort and language-specific investigations.

The SQuAD data set was also evaluated in knowledge base systems, and answers to questions were sought using these systems [14]. They investigated the performance of transformer models pre-trained on the SQuAD dataset and evaluated a dataset created from the knowledge base system along with additional information. The findings demonstrate a substantial improvement compared with the results of the standard models. However, the exact match accuracy score was less than 0.37 [15]. A Turkish question-answering system was developed, enhanced with a knowledge graph, by scraping data from beyazperde.com to respond to inquiries in the movie sector. They evaluated their proposed question-answering pipeline and found that the question-embedding module had a minimal effect on overall performance. Although BiLSTM offers a better representation for 2-hop questions, the SBERT model outperforms it for 3-hop questions [16]. The paper proposes a framework for answering complex questions through question decomposition and web interaction. They released a new dataset, ComplexWebQuestions, featuring questions, SPARQL queries, answers, and web snippets collected using their model [17]. Another study demonstrated that a universal schema serves as a more effective knowledge resource for question-answering than relying solely on a knowledge base or text. Their findings indicate that while a knowledge base is favored over text when it includes relevant facts, a significant number of queries still refer to text, suggesting that a combination of both text and the knowledge base outperforms the knowledge base on its own.

Reference [19] improved short-text classification in Arabic by incorporating external knowledge, such as Wikipedia, to mitigate data sparsity and ambiguity. Semantically enriching the data on knowledge base and also using these data in the attention mechanism with deep learning that have had a positive effect on text classification in the Arabic language. Reference [20] proposed a visual question answering method that integrates image-derived semantic representations with external knowledge base information to address questions beyond the visible content of an image. By combining these textual representations and inputting them into a recurrent neural network alongside the question, the model achieves a more comprehensive scene understanding. This approach enables the answering of complex, natural language questions that require external context.

The knowledge base in our study was created using large language models. Reference [21] study supported the approach by showing that large language models can keep and present many connections within the knowledge base. This study investigates the extent of relational knowledge encoded in pretrained language models without fine-tuning. The analysis reveals that models like BERT can effectively store and recall factual information, performing competitively with traditional NLP methods and supervised baselines in tasks such as cloze-style queries and open-domain question answering. Reference [22] study that enhances knowledge base question answering by incorporating external textual resources to improve key stages of the question answer pipeline. Integrated with a state-of-the-art knowledge base question answering framework, It demonstrates substantial performance gains on the WebQuestions dataset, underscoring the value of combining unstructured text with structured knowledge base data in question answering systems. Reference [23] proposed the integration of structured knowledge bases with unstructured textual data to improve the performance of open-domain question answering within the IBM Watson framework. Structured sources, such as ontologies and curated databases,

[3]State of Art - SQuAD, https://paperswithcode.com/sota/question-answering-on-squad11,2024

[4]BERTurk, https://github.com/stefan-it/turkish-bert, 2024

provide precise and relational representations of factual knowledge, while unstructured sources, including articles, web content, and natural language documents, offer broader contextual information and linguistic diversity. By combining these complementary sources of knowledge, their approach improves Watson's ability to interpret complex queries, retrieve relevant evidence, and generate accurate context-based answers. Reference [24] paper proposed a multicolumn convolutional neural network approach to answer natural language questions over a knowledge base. The model captures question semantics from three perspectives: answer path, answer context, and answer type, while jointly learning low-dimensional embeddings of entities and relations. It was trained using question-answer pairs and enhanced through multitask learning with question paraphrases.

III. THE CREATION OF A DATASET

A. SQUAD DATASET

The Stanford Question Answering Dataset (SQuAD) comprises a collection of triples consisting of content (c), questions (q), and answers (a). This dataset was derived from 10,000 Wikipedia articles, and it was categorized into training and development sets. The training set covered 90% of the articles, whereas the development set contained the remaining 10%. Detailed statistics and sample data related to SQuAD are available in Tables 1 and 2.

TABLE 1. The statistics of English SQuAD.

Item	Number-of-Items and Value
Question	Train:87599 Development:10570
Paragraph	Train:18896 Development:2067
Title	Train:442 Development:48
Human EM Score	86.831
Human F1 Score	89.452
IE-Net EM (Sota) Score	90.939
IE-Net F1 (Sota) Score	93.214

TABLE 2. The sample data of SQuAD.

Item	Value
Title	Nikola Tesla
Paragraph	In 1870, Tesla moved to Karlovac, to attend school at the Higher Real Gymnasium, where he was profoundly influenced by a math teacher Martin Sekulic The classes were held in German, as it was a school within the Austro-Hungarian Military Frontier...
Question	Who was Tesla influenced by while in school?
Answer1	Martin Sekuli
Answer2	Martin Sekuli
Answer3	math teacher Martin Sekuli
Answer Start Point	148

B. THE CREATING OF MSQUAD

We translated the training and development datasets of SQuAD into Turkish using Microsoft Translator and published on GitHub,[5] and we call it mSQuAD (m is coming from Microsoft Translator). As mentioned previously, you can review all the details of this dataset in the article we published. Most operations are summarized here[6] and [1]. After machine translation, the incidence of "content contains answer" was assessed at 62%. This result indicated that nearly half of the dataset may be deemed unusable. To increase the impact of data volume and to work with a larger dataset, we conducted a series of corrective operations aimed at enhancing the usability of the translated output. Furthermore, another dataset, known as gSQuAD (g is from Google Translator) [3], was translated using Google Translator and covers around 50% of SQuAD. gSQuAD served as a reference for comparing and evaluating the respective data sets. This methodology also facilitates the exploration of quality on model accuracy. For this purpose, the content (c), question (q), and answer (a) structures were translated into batches using Microsoft Translator. After identifying and rectifying various translation inaccuracies, we increased the usable size of the dataset to 78%. A sample of the translated dataset is presented in Table 3. By updating the automatic translation service and correcting batch codes for translation errors, we achieved a significant increase in usable data. Compared to the previous study [3] on gSQuAD, 28% more data are available.

TABLE 3. The sample data of mSQuAD.

Item	Value
Title	Nikola Tesla
Paragraph	1870'te Tesla, Yüksek Gerçek Spor Salonu'ndaki okula gitmek için Karlovac'a taşındı ve burada matematik öğretmeni Martin Sekulić'den derinden etkilendi. Dersler Avusturya-Macaristan Askeri Sınırı ...
Question	Tesla'nın Karlovac'taki ana etkisi kimdi?
Answer1	Martin Sekulić
Answer2	Martin Sekulić
Answer3	matematik öğretmeni Martin Sekulić
Answer Start Point	128

C. POST TRANSLATION ACTIONS

Post-translation analysis showed approximately 60% of the data were in correct triple form (content(c)-answer(a)-questions(q)) for model training and testing. Microsoft Translator provided 10% more usable data than Google Translate. To minimize data loss, error cases were reviewed, and the lost data were reintroduced into the usable dataset using fixed-batch data codes.

Case 1: Basic syntax errors appeared in translation outputs, with extraneous characters such as dots, commas, and spaces (e.g., en: "alternatingcurrent", tr: "alternatif akm.") and various encoding problems (e.g., '–,-', content: "14-12",

[5] https://github.com/kadirtutar/datasetTranslated
[6] UBMK, 9th International Conference on Computer Science and Engineering, https://ubmk.org.tr/en/

TABLE 4. Increases obtained with manual and batch code-based corrections after machine translation.

Case-Number	Devevelopment-Dataset-Rate	Train-Dataset-Rate
Case 1	+%5	+%3
Case 2	+%6	+%5
Case 3	+%4	+%4
Case 4	+%1	+%1

answer: "14-12"). Date format errors were common; for instance, en-answer: "February 7, 2016" tr-answer: "7 Şubat 2016, Şubat 2016, Bugün", correct answer: "7 Şubat 2016".

Case 2: All data were converted to lowercase, but this was not sufficient for matching tasks. Turkish documents may have specific English terms and unique casings (e.g., Turkish: I,İ,ı,i vs. English: I,i). Exact matches emerged after adjusting for casing and replacing language-specific characters. Casing issues were resolved using custom methods for uncapitalizing and capitalizing.

Case 3: Some English proper nouns are erroneously translated into Turkish. When original English answers were present in Turkish content, translations were updated to match; for instance, "Arizona Cardinals" should remain unchanged instead of being translated as "Arizona Kardinalleri." Additionally, "The" was removed from English answers during comparison. These adjustments added data to the dataset.

Case 4: Discrepancies arise when translating monetary values, numbers, and mathematical notations. For example, content might show "1,200" while the answer shows "1.200," or "10%" might translate to "%10." Similarly, "5,000,000$" may appear as "5.000.000 dollar." Corrections ensured consistency between context and answers, with inappropriate characters or terms removed or replaced in the batch script. The effect of these corrections on the size of the dataset is in Table 4.

By updating the automatic translation service and correcting batch codes for translation errors, we achieved a significant increase in usable data.

IV. CREATING BASELINES & RESULTS

A. BERT (BIDIRECTIONAL ENCODER REPRESENTATIONS TRANSFORMERS) BASELINE

In recent years, transformer-based models have been used in question-answering tasks owing to their robust architecture. BERT is a key model. Upon its introduction, BERT achieved state-of-the-art accuracy in various NLP tasks, including language understanding and question answering. BERT uses a semi-supervised learning approach. First, the model was trained on a large corpus, such as Wikipedia, to grasp language patterns. Next, the fine-tuning process adapts BERT to specific tasks. BERT comprises a transformer architecture encoder-decoder stack that employs self-attention, enabling better understanding and interpretation of context.

As training a large language model is costly and not the focus of the study, the current BERTurk[7] large language model was used in this study. The current BERTurk version was trained on a filtered, sentence-segmented Turkish OSCAR corpus, a recent Wikipedia dump, various OPUS corpora, and a special corpus. Its corpus size is 35GB with 4,404,976,662 tokens. In the context of the BERT infrastructure for question and answer tasks, we established a baseline for fine-tuning using our mSQuAD dataset. It was called the BERT baseline. The process involves feeding the mSQuAD dataset with the BERTurk model for fine-tuning. The output of this fine-tuning step is a new model called BERT-Baseline-Model-TR.

B. OPENAI CHATGPT & QWEN2 BASELINES

OpenAI is an AI research organization founded in December 2015 and based in San Francisco, California. Its mission is to develop "safe and beneficial" artificial general intelligence (AGI), defined as "highly autonomous systems that outperform humans at most valuable work." As a leader in the AI boom, OpenAI belongs the GPT family of language models. The release of ChatGPT in November 2022 sparked widespread interest in generative AI. Two versions of ChatGPT (ChatGPT 3.5 Turbo and ChatGPT 4.1) were added to the system as baselines.

In addition, the open-source Qwen2 (Qwen/Qwen2-7B-Instruct)[8] language model has been added as a baseline to enrich the system for comparison and evaluation. Qwen2 represents the latest generation in the Qwen family of large language models. The present repository specifically provides access to the instruction-tuned Qwen2 model with 7 billion parameters. In comprehensive benchmark evaluations covering tasks such as language understanding, text generation, multilingual processing, programming, mathematics, and logical reasoning, Qwen2 has consistently outperformed most existing open-source models, including its predecessor Qwen1.5. Moreover, it demonstrates competitive performance relative to leading proprietary models, thereby establishing Qwen2 as a state-of-the-art open-source alternative for a wide range of natural language processing applications.

All these models serve in the Generative AI architecture together with large language models. Generative artificial intelligence (GAI) is a subset of artificial intelligence that uses generative models to produce text, images, videos, or other data. These models learn the patterns and structures of their training data and use them to create new data based on inputs, often in the form of natural language prompts. Prompt engineering involves structuring an instruction understood by a generative AI model. A prompt is a natural language text that describes AI tasks. A prompt for a text-to-text model can be a query like "What did Newton discover?", a command like "Can you write a poem that describes love like Sheaskper?", or longer statements with context, instructions, or conversation history.

[7]BERTurk, https://github.com/stefan-it/turkish-bert

[8]Qwen2, https://huggingface.co/Qwen/Qwen2-7B-Instruct

The ChatGPT generative models that we use with the OpenAI infrastructure usually need to receive strong prompts from outside to reach the complete answer. This requirement also applies to the Qwen2-based system. SQuAD tasks expect a specific answer within a paragraph to be a word or group of words. However, there are two fundamental problems for ChatGPT and Qwen2. The first problem is that it should be addressed by considering the given paragraph. Generative artificial intelligence has a natural structure that tends to produce responses from training data independently of inputs. For the system to respond depending on the paragraphs, it must be directed promptly. The second problem is that the system should provide the exact answer instead of the closest sentence while answering. These rules can also be implemented with prompt guidance. To address these problems, ChatGPT & Qwen2 is given a question, a paragraph, and two prompts. The first prompt allows answers to be given only by sticking to the paragraph. The second prompt allows the answer to be given precisely through words instead of similar sentences. The prompts used are listed in Table 5. The content, questions, and prompts are fed into two versions of the ChatGPT models (ChatGPT 3.5 Turbo and ChatGPT 4.1) and Qwen2 in the architecture of baselines system. Questions are also directed at the models, combining with the other inputs to produce meaningful outputs.

TABLE 5. Baselines prompt samples.

	Applied Prompts
Prompt-1	You must answer the following question, taking into account the data in the paragraph.
Prompt-2	The answer should be given in words only, not in sentences.

C. BASELINE RESULTS

The first baseline model was built using BERTurk with a deep learning infrastructure. This model was fine-tuned on mSQuAD. After creating the Bert baseline, two additional ChatGPT-based and one Qwen2 comparison baselines were created. One of them is the ChatGPT 3.5 turbo, and another model is the ChatGPT 4.1 (release date: gpt-4.1-2025-04-14), which is one of the most successful and up-to-date models in this field. Qwen2 is one of the most successful open source models. By running four baseline models, the results were compared and analyzed according to their exact match (EM) and F1 scores. The exact match and F1 scores of the four baseline models are listed in Table 6. While the BERT baseline model is 4% more successful than the ChatGPT 3.5 baseline model for exact matching, the results for the F1 score are very close to each other. As expected, ChatGPT 4.1 produced the highest exact match and F1 scores compared to the other models (EM:+8, F1:+10). The performance of the Qwen2 model was lower than all baseline models.

Baseline results were examined in detail from different perspectives. Factors affecting model performance were identified. The length of the data is a crucial factor that influences the effectiveness of a model. Model success must act stably regardless of the length of data given to the system. The length of the paragraph, question, or answer might affect the success of the system. We observed that extended input lengths negatively affected the overall success of the models. Analyzing the impact of content length (c), questions (q), and answers (a) revealed that longer inputs negatively affect model success. There is a significant correlation between input length and stability. We calculated accuracy values based on average token limits for each input type, with results shown in Table 7.

TABLE 6. Results of baseline models on mSQuAD.

Model	EM	F1
Bert Baseline Model (B1)	0.49	0.57
ChatGPT 3.5 Baseline Model (B2)	0.45	0.56
ChatGPT 4.1 Baseline Model (B3)	0.57	0.67
Qwen2-7B-Instruct Baseline Model (B4)	0.34	0.44

In addition to the length of the data, its content is significant. Models perform better with general content such as places, cities, and biographies, but face challenges with specific terminology. In particular, running on articles where technical terms, abstract concepts, scientific terms, art, and philosophy are discussed. In such cases, models struggle to determine the answers. The details of these findings are in Table 8. While creating Table 8, we took some samples of the most successful and a few of the unsuccessful for BERT. "Baseline1 Rank" column shows the success order of BERT between domains (the 5 most successful domains vs the 5 least successful domains). Based on these values, the successes of the ChatGPT and Qwen2 baselines were compared domain by domain.

Model success also varied depending on the type of question asked. They performed well with simple questions like "when," "how much," and "who," but struggled with open-ended ones like "which" and "what". Detailed results for these question types are in Table 9.

In the studies conducted on the baseline models, it was observed that the domain-based accuracy values of the models changed, and that the accuracy values changed according to the length of the given data and the length of the question. In addition, the accuracy values changed according to each question type. Therefore, it is extremely important to ensure that the models have a balanced structure according to changing data content, questions, and lengths.

V. EXPERIMENTS WITH STRUCTURED AND ENRICHED DATA

A. CREATING STRUCTURED AND ENRICHED DATA

1) CREATING STRUCTURED PARAGRAPH DATA

When we analyzed the results of the baselines, we found that the results varied depending on the given paragraph or question. It is difficult to address this variability and problems in LLM, transformer-based architecture, and ChatGPT Saas

TABLE 7. BERT, ChatGPT, Qwen2 baselines with token-count (c = content, q = question, a = answer), Baseline = B, Token Size = #, Token Type = T.

B1	EM	F1	B2	EM	F1	B3	EM	F1	B4	EM	F1	#	T
BERT	0.54	0.62	GPT3.5	0.44	0.56	GPT4.1	0.57	0.67	Qwen2	0.33	0.44	< 95	c
BERT	0.45	0.51	GPT3.5	0.45	0.56	GPT4.1	0.58	0.68	Qwen2	0.34	0.44	≥ 95	c
BERT	0.52	0.60	GPT3.5	0.54	0.57	GPT4.1	0.58	0.66	Qwen2	0.32	0.43	< 7	q
BERT	0.49	0.56	GPT3.5	0.44	0.56	GPT4.1	0.57	0.68	Qwen2	0.34	0.44	≥ 7	q
BERT	0.54	0.60	GPT3.5	0.47	0.58	GPT4.1	0.63	0.73	Qwen2	0.34	0.44	$= 1$	a
BERT	0.43	0.52	GPT3.5	0.49	0.60	GPT4.1	0.64	0.75	Qwen2	0.33	0.43	$= 2$	a
BERT	0.37	0.47	GPT3.5	0.42	0.57	GPT4.1	0.55	0.65	Qwen2	0.34	0.45	$= 3$	a

TABLE 8. Domain based evaluation - BERT, ChatGPT, Qwen2 Baselines, Baseline = B.

Domain	B1	EM	F1	B2	EM	F1	B3	EM	F1	B4	EM	F1	B1 Rank
Islamism	BERT	0,72	0,77	GPT3.5	0,38	0,51	GPT4.1	0,51	0,62	Qwen2	0,31	0,41	1
1973 oil crisis	BERT	0.68	0.74	GPT3.5	0.48	0.60	GPT4.1	0,61	0,66	Qwen2	0,45	0,53	2
Kenya	BERT	0.66	0.76	GPT3.5	0.43	0.56	GPT4.1	0.44	0.54	Qwen2	0,28	0,43	3
Harvard University	BERT	0.63	0.63	GPT3.5	0.61	0.71	GPT4.1	0.66	0.73	Qwen2	0,51	0,57	4
Sky (United Kingdom)	BERT	0.62	0.66	GPT3.5	0.50	0.62	GPT4.1	0.65	0.74	Qwen2	0,36	0,43	5
Civil disobedience	BERT	0.38	0.47	GPT3.5	0.44	0.50	GPT4.1	0.46	0.57	Qwen2	0,34	0,40	44
European Union law	BERT	0.36	0.53	GPT3.5	0.36	0.52	GPT4.1	0.52	0.62	Qwen2	0,26	0,42	45
Prime number	BERT	0.35	0.43	GPT3.5	0.28	0.40	GPT4.1	0.47	0.59	Qwen2	0,23	0,38	46
Chlorop- last	BERT	0.34	0.48	GPT3.5	0.26	0.38	GPT4.1	0.38	0.51	Qwen2	0,18	0,32	47
Computa- tional theory	BERT	0.34	0.51	GPT3.5	0.26	0.38	GPT4.1	0.42	0.54	Qwen2	0,14	0,25	48

TABLE 9. BERT, ChatGPT, Qwen2 baselines with question type, samples from question type count ≥ 100, Baseline = B.

Question Type	B1	EM	F1	B2	EM	F1	B3	EM	F1	B4	EM	F1	B1 Rank
Ne zaman (When)	BERT	0,76	0,80	GPT3.5	0.72	0.60	GPT4.1	0.77	0.85	Qwen2	0,27	0,43	1
Kaç (How much)	BERT	0.69	0.73	GPT3.5	0.48	0.33	GPT4.1	0.47	0.63	Qwen2	0,24	0,33	2
Ne kadar (How much)	BERT	0.62	0.67	GPT3.5	0.49	0.34	GPT4.1	0.46	0.63	Qwen2	0,19	0,31	3
Kim (Who)	BERT	0.57	0.63	GPT3.5	0.72	0.62	GPT4.1	0.73	0.79	Qwen2	0,57	0,65	4
Nerede (Where)	BERT	0.45	0.52	GPT3.5	0.55	0.43	GPT4.1	0.51	0.63	Qwen2	0,19	0,34	5
Hangi (Which)	BERT	0.51	0.59	GPT3.5	0.65	0.54	GPT4.1	0.65	0.73	Qwen2	0,46	0,56	6
Ne (What)	BERT	0.42	0.52	GPT3.5	0.48	0.36	GPT4.1	0.50	0.61	Qwen2	0,28	0,39	7

infrastructure or Qwen2. Large language models (BERTurk, Qwen2), transformer-based models (BERT), and SAAS services (OpenAI, ChatGPT) are generally large black box systems. We observed that regardless of the model structures, making sense of the data, enriching it, and transforming it into a structural form changed the model behavior. Therefore, we focused on making the data more structured.

Normally, when humans interpret a language, we divide the sentences into words and interpret the relationships between these words to reach the meaning of the sentence. If we need to store this structure per the computer data structure, words in sentences can be presented in a graph structure as a node or link. The structural storage of the domain data in this structure is called a knowledge graph. A knowledge graph consists of nodes and links. Relationships in the triple structure are structurally maintained as subject-predicate-objects. While subjects and objects are stored on nodes, predicates are stored on links.

There are different methods for building a knowledge base. Each method has its advantages and disadvantages [25]. It can be evaluated using various KPIs such as the type of problem being solved, coverage, resources, time, and accuracy. Some knowledge base creation methods are listed in Table 10 with KPIs, properties, and their effective features are listed below.

Considering all the KPIs and objectives of the study, the most feasible method was to utilize an LLM-assisted knowledge creation approach.

- **LLM-Assisted:** Excellent for fast, broad KB population using external knowledge, though it requires careful oversight to mitigate hallucinations.
- **Manual Creation:** Highest accuracy and reliability, but impractical for scaling and incurs high maintenance and human effort costs.
- **IR-Based Methods:** Highly scalable and efficient for coverage-oriented tasks.
- **Traditional NLP Pipelines:** Especially effective in domain-specific contexts with structured extraction needs.

TABLE 10. Evaluation of knowledge base creation technique, ranking(1 to 10, Low:0-3, Medium:4-7, High:8-10).

Technique	Accuracy	Coverage	Human Effort	System Cost
LLM-Assisted	High (9)	High(9)	Low(3)	Medium(6)
Manual Creation	High (10)	Low(3)	High (10)	Low(2)
IR-Based Meth- ods	Medium(7)	Medium(7)	Medium(5)	Medium(5)
Traditional NLP Pipelines	High (8)	Medium(7)	Medium(5)	Medium(5)

ChatGPT, OpenAI, and Langchain Language have released a pipeline that will accelerate the solution to this obstacle. They provide an infrastructure for transforming unstructured data into structural data according to the data scheme. Using this infrastructure, we created knowledge graphs from the unstructured paragraph data. The ChatGPT language model can understand the important entities in these data and can determine the relationships between these entities. Langchain libraries transform these outputs into structural forms in a graph knowledge model.

We explain in detail how structured data are formed in Figure 1. Paragraph data were fed into the system to convert it into a structured format. In addition, we provide our graph schema data using prompts. Additionally, we used the Langchain[9] and ChatGPT infrastructure to evaluate these inputs and receive outputs from the unstructured data. Langchain is a composable framework for building with LLMs and an orchestration framework for controllable agentic workflows. Then we stored the structured data in the graph database (Neo4J database[10]). Neo4j is a graph database management system (GDBMS) developed by Neo4j, Inc.

The manner in which paragraph data are transformed into structural form is shown in Table 11. A paragraph entering the system is first processed and transformed with ChatGPT, LangChain and structural schema structures like ''Output'' column in Table 11 and Figure 1. This graph data consists of many triples, which were stored in Neo4J. During the training and testing phase, these triples data are converted into a paragraph in Table 11 as ''Output'' column for ''Knowledge Base Graph Data'' row.

2) CREATING ENRICHED QUESTION DATA

After the paragraph was structured in the pipeline, we carried out an enriched question data process. To enrich the question data, the effectiveness and success of different methods were first evaluated. The accuracy effects of question enrichment were measured on Baseline1. Three options were considered: Option-1 enriched the question using a prompt call on ChatGPT. Option-2 paraphrased questions on T5.[11] Option-3 generated semantic similarity using the Turkish model of Fasttext[12] (cc.tr.300.vec) using the Gensim library. The results were obtained using different techniques to ask questions differently on the Baseline1 model. The results were evaluated based on the effects of the original question asked in the Baseline1 model and the effects of the question being modified according to different techniques in Table 12. It was decided that the most successful method would be to use the prompt call technique on ChatGPT.

To determine the impact of the prompts used on the output and accuracy, four alternative prompts were created with different words, but their meanings are the same. These prompts were tested on a Baseline1 model. The accuracy effects are shown in the Table 13. A negligible effect of $+1/-1\%$ was observed with rounding.

Studies have been conducted to identify the important parts of the question from different perspectives. We collected data from these studies using three main items. The first item is to determine how the question can be asked differently with the help of prompts. The second item determines which object the question focuses on, and the third item determines what the sought object is related to in the question.

- Alternative Question: Describes how we might ask this question differently.
- Answer Related Object: Which objects in the question could possibly be related to the answer.
- Important Object: Which is the most important ojbect in the question of finding an answer.

We concentrated on addressing these issues using ChatGPT and the Langchain framework. We received three outputs based on three distinct prompts applied to the question. The study continued with the prompts specified in Table 14. In addition to the original question, three different sets of enriched question data were also stored in the database together with the original question. The samples of enriched question data for different questions are listed in Table 15.

B. TRAIN WITH STRUCTURE DATA

A new model was developed through the meticulous process of refining the existing BERT architecture, using the mSQuAD dataset. The architecture was fed into using ''Formatted KnowledgeBase Content Data'' and ''Enriched Question Data''. Unlike baselines model training, in the current architecture, knowledge-based structured paragraph data was fed into the system in addition to the original data. In addition to the original question, the enriched question data obtained was fed into the system. Enriched question data in the form of how to ask the question, alternatively, the relevant part of the question, and the most important part of the question were additionally fed into the system while training the model. A detailed description of the components and flow is provided in the following.

- BertTR: The pre-trained Turkish large language model.
- Training Data:
  - mSQuAD: Translated SQuAD dataset into Turkish.
  - Formatted KnowledgeBase Content Data: Structured paragraph data from coming knowledge base database.
  - Enriched Question Data: alternative question + answer related object + important object
- Fine-Tuning Process: BERT infrastructure
- Output (New BERT-Model-TR): New KnowledgeBase Model

The effect of data enrichment and structuring on accuracy is examined in detail in the Results section.

[9]Langchain, https://www.langchain.com/
[10]Neo4J, https://neo4j.com/
[11]T5 Multilingual Model mT5 multilingual XLSum, https://huggingface.co/csebuetnlp/
[12]Meta Fasttext, https://ai.meta.com/tools/fasttext/

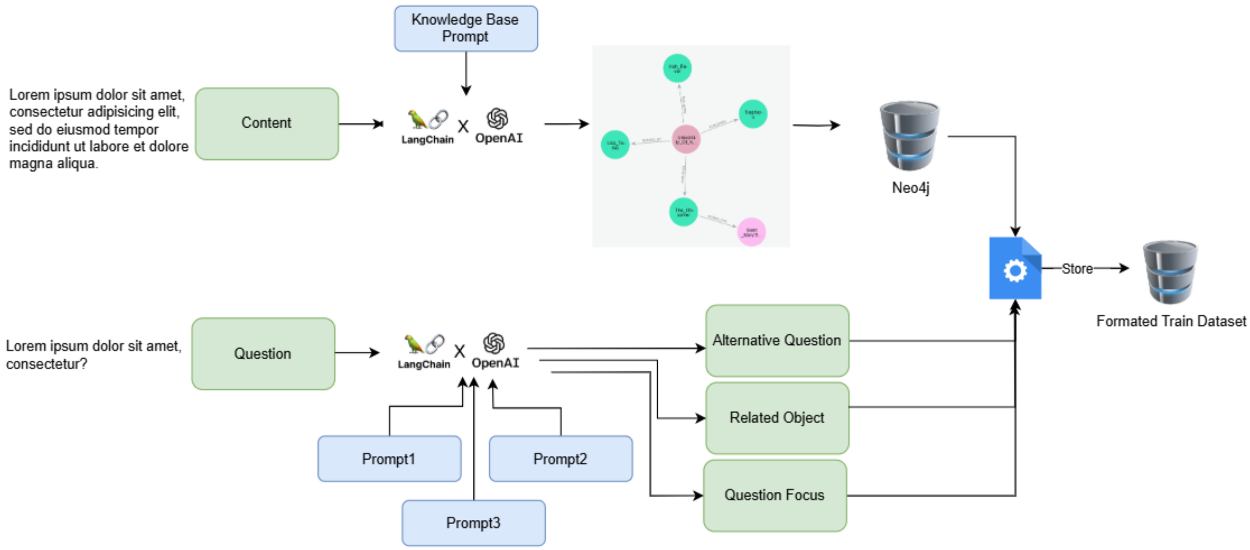


FIGURE 1. **Structure data transformation process.**

C. RESULTS

In our evaluation of the baselines, we observed that the success of the model varies according to the content length, between domains, and question types. In this section, the outputs of the newly created structure-based trained model (KnowledgeBase model) are evaluated. The success of the new model was based on the following main headings:

- Overall results
- Results by content length
- Results by domain
- Results by question type

When the model was trained in different combinations with structural data, it was observed that the system trained with structure paragraph data and enriched question data provided significant superiority in terms of both F1 score and EM score compared to the previously developed baseline models. The success of different models trained with baseline results, structured paragraph data, and enriched question data is shown in Table 16 and Figure 2.

Two different methods were applied when training the model with structured paragraph and enriched question data. In the first model, the model was trained with only enriched question data, and in the second model, structural paragraph data were also added. The highest performance was achieved in the structure of the second model. Compared with the baseline of BERT, 11% more precision was achieved for both EM and F1. Compared with the ChatGPT 3.5 baseline, a 15% increase in accuracy was achieved for the EM and 12% F1 scores. It also produced 3% better results in EM and 1% better results in F1 scores than the extremely powerful ChatGPT 4.1 model. Compared with Qwen2 baseline, a 25% increase for the EM and F1 scores.

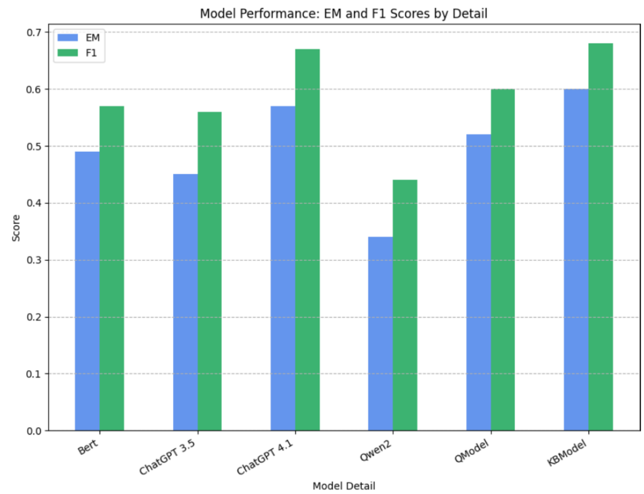


FIGURE 2. **Performance of BERT baseline, ChatGPT baselines, Qwen2 baseline, QModel (Question Enrichment), KBModel (Structured Content + Question Enrichment).**

In summary, baseline1 (BERT) performs slightly better than baseline 2 (ChatGPT 3.5) for both EM and F1 scores. The baseline 4 based on Qwen2, generally produced lower results than all other models. It was not considered a competitive baseline in the error analysis. QModel (Enrichment Questions) improves the performance of both baselines. However, baseline 3 (Chatgpt 4.1) produced better results than the QModel for both F1 and EM scores. KBModel (Structured Data + Enrichment Questions) significantly outperforms all other models, suggesting that combining structured content with enriched questions leads to the best performance.

TABLE 11. Transformation of structured data sample.

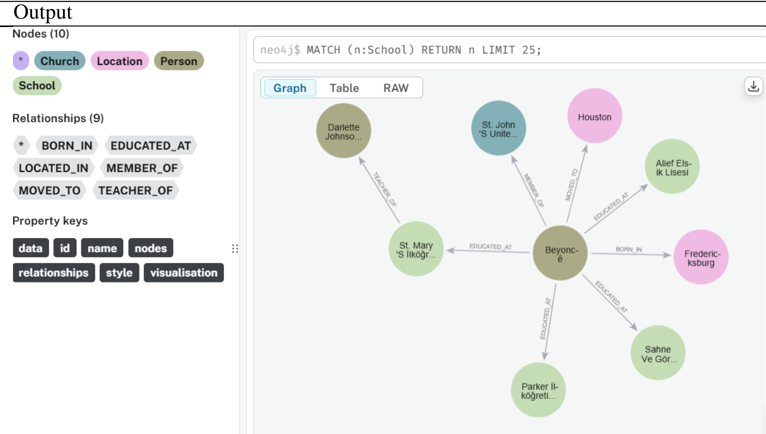
Data	Output
Beyoncé, Fredericksburg, Teksas'taki St. Mary's İlköğretim Okulu'na gitti ve burada dans derslerine kaydoldu. Şarkı söyleme yeteneği, dans eğitmeni Darlette Johnson'ın bir şarkıyı mırıldanmaya başlamasıyla keşfedildi ve onu bitirdi, tiz notalara ulaşabildi. Beyoncé'nin müziğe ve performansa olan ilgisi, yedi yaşında john Lennon'ın "Imagine" şarkısını söyleyerek 15/16 yaşındakileri yenmek için bir okul yetenek gösterisini kazandıktan sonra da devam etti. 1990 sonbaharında Beyoncé, Houston'daki bir müzik mıknatısı okulu olan Parker İlköğretim Okulu'na kaydoldu ve burada okulun korosuyla birlikte sahne alacaktı. Ayrıca Sahne ve Görsel Sanatlar Lisesi'ne ve daha sonra Alief Elsik Lisesi'ne gitti. Beyoncé ayrıca st. John's United Methodist Kilisesi'nde iki yıl boyunca solist olarak koronun bir üyesiydi.	
Knowledge Base Graph Data	Beyoncé, St. Mary'S İlköğretim Okulu'na gitti. Beyoncé, Darlette Johnson'ı keşfetti. Beyoncé, John Lennon şarkısını seslendirdi. Beyoncé, Parker İlköğretim Okulu'na gitti. Beyoncé, Sahne Ve Görsel Sanatlar Lisesi'ne gitti. Beyoncé, Alief Elsik Lisesi'ne gitti. Beyoncé üyesi St. John'S United Methodist Kilisesi.

TABLE 12. Results of question enrichment trial methods.

Approach	EM	F1	Result
Bert Baseline Model	0.49	0.57	-
Prompt Calling (ChatGPT)	0.51	0.57	Partial better performance
Word2Vec Similarity (fastText)	0.41	0.50	Less accuracy
Paraphrasing (mT5)	0.11	0.12	Unsuccessful

TABLE 13. Accuracy of different prompts on Baseline 1 for A alternative question.

Applied Prompt	F1	EM
Indicates that it describes how we might ask this question differently.	0.49	0.57
Suggests a way to rephrase the question to convey the same meaning.	0.49	0.57
Provides a different wording or structure for asking the same question.	0.48	0.56
Offers an alternative formulation of the original question while preserving intent.	0.50	0.56

1) RESULTS BY CONTENT LENGTH

Table 17 compares the KB model with three baselines in terms of performance metrics, EM, and F1 scores. Data were categorized according to token count perspectives (content, question, answer) and thresholds such as <95 for content, <7 for question, = 1, = 2, and = 3 for the answer.

TABLE 14. Baseline ChatGPT prompt structure.

Type	Applied Structure Data&Prompts
Alternative Question	Indicates that it describes how we might ask this question differently.
Answer Related Object	Indicates that which objects in the question could possibly be related to the answer?
Important Object	Indicates that which is the most important ojbect in the question to find the answer?

Content (c), Token Range: < 95: KB Model achieves the best performance, improving **EM by 6%** and **F1 by 6%** compared to Baseline1. It also increased the EM score by 3% and F1 score by 1% compared with the most successful Baseline 3 (ChatGPT 4.1) model.

Content (c), Token Range: >= 95: KB Model achieves the best performance, improving **EM by 13%** and **F1 by 15%** compared to Baseline1. When we compare according to Baseline 3, while the EM match scores are the same, the F1 score is 2% lower.

Question (c), Token Range: < 7: KB Model achieves the best performance, improving **EM by 9%** and **F1 by 8%** compared to Baseline1. It also increased the EM score by 3% and the F1 score by 2% compared with Baseline 3.

Question (c), Token Range: >= 7: KB Model achieves the best performance, improving **EM by 9%** and **F1 by 9%** compared to Baseline1. When we compare according to

TABLE 15. Sample of enriched question data.

Question	Alternative Question	Answer Related Object	Important Object
Notre Dame Scholastic Dergisi ne zaman yayınlanmaya başladı?	Notre Dame Scholastic Dergisi'nin ilk yayın tarihi ne zaman?	yayınlanma zamanı	Notre Dame Scholastic Dergisi
Notre Dame Ana Binası'nın önünde ne var?	Notre Dame Ana Binası'nın ön cephesinde yer alan yapı nedir?	önünde	Notre Dame Ana Binası
Notre Dame'daki Mühendislik Koleji hangi yılda kuruldu?	Notre Dame Üniversitesi Mühendislik Fakültesi'nin kuruluş yılı nedir?	kuruluş yılı	Mühendislik Koleji
Mendoza College of Business'daki lisans okulu BusinessWeek'e göre nerede sıralandı?	Mendoza College of Business'ın BusinessWeek sıralamasındaki yeri nedir?	lisans okulu	Business Week

TABLE 16. Performance of BERT Baseline, ChatGPT Baselines, QModel (Question Enrichment), KBModel (Structured Content + Question Enrichment).

Model	Detail	EM	F1
Baseline1	Bert baseline model	0.49	0.57
Baseline2	ChatGPT 3.5 baseline model	0.45	0.56
Baseline3	ChatGPT 4.1 baseline model	0.57	0.67
Baseline4	Qwen2 baseline model	0.34	0.44
QModel	Model with enrichment question	0.52	0.60
KBModel	Model with content structure data + enrichment question (Call as knowledgeBase model)	0.60	0.68

Baseline 3, while the EM is better than +%1, the F1 score is 3% lower.

Answer (c), Token Range: = 1: KB Model achieves the best performance, improving **EM by 13%** and **F1 by 11%** compared to Baseline1. It also increased the EM score by 4% and the F1 score by 2%.

Answer (c), Token Range: = 2: KB Model achieves the best performance, improving **EM by 18%** and **F1 by 17%** compared to Baseline1. Compared with Baseline 3, the EM score showed a 3% lower performance, and the F1 score showed a 6% lower performance.

Answer (c), Token Range: = 3: KB Model achieves the best performance, improving **EM by 17%** and **F1 by 17%** compared to Baseline1. Compared with Baseline 3, the EM score showed a 1% lower performance, and the F1 score showed a 2% lower performance.

KB Model outperforms both baselines significantly, achieving **an EM improvement of 6-18%** and an **F1 improvement of 6-17%**. The KB Model consistently outperformed both Baseline1 (BERT) and Baseline2 (ChatGPT) across all token ranges and types, particularly in answer-focused tasks. The performance of all models varies across token ranges; however, the KB Model maintains its superiority in handling both short and long token ranges effectively. If we evaluate it in terms of token count according to Baseline3, KB Model has achieved more successful results in short texts and exact match scores.

2) RESULTS BY DOMAIN

The Table 18 compares the performance of five models: Baseline1 (BERT), Baseline2 (ChatGPT 3.5), Baseline3 (ChatGPT 4.1), Baseline4(Qwen2) and KB model in various domains using two evaluation metrics: exact match and F1 score. The order in Table 18 is created according to the domains in where baseline1 is the most successful and the least successful. Against these domains, the KB model achieved more successful results in many domains than in all the baselines. In the domains where Baseline 1 and Baseline 3 were successful (Rank 1-5), the KB model achieved the most successful results. In addition, the KB model achieved more success in the domains where Baseline 1 and Baseline 3 were the least successful.

Baseline1 Top 5 Domain : KB Model achieves the best performance, improving **EM by 8%** and **F1 by 14%** compared to Baseline1. It also achieves the best performance, improving **EM by 18%** and **F1 by 17%** compared to Baseline3.

Baseline1 Down 5 Domain : KB Model achieves the best performance, improving **EM by 18%** and **F1 by 11%** compared to Baseline1. It also achieves the best performance, improving **EM by 14%** and **F1 by 10%** compared to Baseline3.

The performance of all models varies domain-by-domain, but The KB model offers more successful and balanced results. When we evaluated ChatGPT 3.5 and ChatGPT 4.1, we observed no consistent improvement as far as the number of tokens was concerned, and in some cases, ChatGPT 3.5 also produced more accurate results than ChatGPT 4.1.

3) RESULTS BY QUESTION TYPE

Table 19 evaluates the performance of the five Baseline1 (BERT), Baseline2 (ChatGPT 3.5), Baseline3 (ChatGPT 4.1), Baseline4(Qwen2) and KB models across various types of questions using the following metrics: exact match and F1 score. Table 19 was created according to the seven types of questions (number of each type > 100), with ranking of Baseline1 being the most successful. The success of the KB model was evaluated against the 7 types of questions.

- **Nc Zaman (When):**
  - BERT performs moderately EM (0.76) and F1 (0.80).
  - KB Model achieves the highest EM (0.81) but is lower in F1 (0.75).
  - ChatGPT 3.5 performs moderately (EM: 0.72, F1: 0.60).
  - ChatGPT 4.1 achieves the highest F1 (EM: 0.75, F1: 0.85).

TABLE 17. KB model vs baselines with tokenize count perspective as c = content, q = question, a = answer.

B1	EM	F1	B2	EM	F1	B3	EM	F1	B4	EM	F1	Model	EM	F1	#Token	Type
Bert	0.54	0.62	GPT 3.5	0.44	0.56	GPT 4.1	0.57	0.67	Qwen2	0.33	0.44	KB Model	0.60	0.68	< 95	c
Bert	0.45	0.51	GPT 3.5	0.45	0.56	GPT 4.1	0.58	0.68	Qwen2	0.34	0.44	KB Model	0.58	0.66	≥ 95	c
Bert	0.52	0.60	GPT 3.5	0.54	0.57	GPT 4.1	0.58	0.66	Qwen2	0.32	0.43	KB Model	0.61	0.68	< 7	q
Bert	0.49	0.56	GPT 3.5	0.44	0.56	GPT 4.1	0.57	0.68	Qwen2	0.34	0.44	KB Model	0.58	0.65	≥ 7	q
Bert	0.54	0.60	GPT 3.5	0.47	0.58	GPT 4.1	0.63	0.73	Qwen2	0.34	0.44	KB Model	0.67	0.71	$= 1$	a
Bert	0.43	0.52	GPT 3.5	0.49	0.60	GPT 4.1	0.64	0.75	Qwen2	0.33	0.43	KB Model	0.61	0.69	$= 2$	a
Bert	0.37	0.47	GPT 3.5	0.42	0.57	GPT 4.1	0.55	0.66	Qwen2	0.34	0.45	KB Model	0.54	0.64	$= 3$	a

- **Kaç (How Much):**
  - BERT follows with EM (0.69) and F1 (0.73).
  - KB Model achieves the highest scores (EM: 0.73, F1: 0.75).
  - ChatGPT 3.5 lags significantly (EM: 0.48, F1: 0.33).
  - ChatGPT 4.1 lags significantly (EM: 0.47, F1: 0.63).
- **Ne Kadar (How Many):**
  - BERT scores EM (0.62) and F1 (0.67).
  - KB Model follows with EM (0.57) and F1 (0.65).
  - ChatGPT 3.5 struggles with EM (0.49) and F1 (0.34).
  - ChatGPT 4.1 struggles with EM (0.46) and F1 (0.63).
- **Kim (Who):**
  - BERT scores EM (0.57) and F1 (0.63).
  - KB Model follows ChatGPT 4.1 with small differences (EM: 0.72, F1: 0.77).
  - ChatGPT 3.5 scores are lower than all models (EM: 0.55, F1: 0.43).
  - ChatGPT 4.1 scores are the most successful (EM: 0.73, F1: 0.79).
- **Nerede (Where):**
  - BERT scores EM (0.45) and F1 (0.52).
  - KB Model achieves the highest EM (EM: 0.64, F1: 0.58).
  - ChatGPT 3.5 performs poorly (EM: 0.43, F1: 0.35).
  - ChatGPT 4.1 achieves the highest F1 (EM: 0.51, F1: 0.63).
- **Hangi (Which):**
  - BERT scores EM (0.51) and F1 (0.59).
  - KB Model performs as ChatGPT 4.1 for EM (0.65) and F1 (0.62).
  - ChatGPT 3.5 lags with EM (0.54) and F1 (0.46).
  - ChatGPT 4.1 achieves the highest F1 (0.65) and F1 (0.73).
- **Ne (What):**
  - BERT achieves EM (0.42) and F1 (0.52).
  - KB Model achieves the highest EM (0.61) and F1 (0.53).
  - ChatGPT 3.5 has a similar EM (0.48) but lower F1 (0.36).
  - ChatGPT 4.1 achieves the highest F1 (0.50) but lower F1 (0.61).

Overall Performance: KB Model consistently performs well across question types for exact match, and surpasses Baseline1, Baseline2 and Baseline4. KB Model excels particularly in EM scores for "Kim (Who)" and "Nerede (Where)."

D. ERROR ANALYSIS

An in-depth error analysis involves the examination of error results from various perspectives. This includes a thorough investigation of the question structure, expected responses, methodologies of BERT-based models, and behavior patterns of ChatGPT (generative AI), particularly through failure cases. The primary focus of this study and the dataset was a question-answering system for reading comprehension. Given that this study focused on reading comprehension in Turkish, error analyses were interpreted within the context of this task. Within the test set, instances of model failure were meticulously examined individually, allowing the categorization of certain errors into specific groups. Observations of these errors have been documented. The BERT model utilizes span classification, whereas ChatGPT operates on a generative model framework, which can impact results in both positive and negative ways. Before conducting a comprehensive analysis, it is essential to provide a general evaluation of the two models. The Qwen2 model produced less accuracy than all other models so this model was not detailed in the error analysis study.

- In terms of performance, the BERT model demonstrates a significantly faster response time.
- Although the precision of the BERT model (KBModel) is higher than other baselines. GPT 3.5 results tend to be more unpredictable and varied than GPT 4.1.
- Error interpretation and detection are more straightforward within BERT base models, whereas this process is more complex with ChatGPT.
- In particular, BERT models do not exhibit the risk of hallucination, while instances of hallucinations have been identified in failure cases involving ChatGPT.

Errors can be classified into several categories, including those specific to the Turkish language, difficulties associated with grammatical structures, the impacts of punctuation and spelling, and variations arising from different modeling

TABLE 18. KB model vs baselines - Domain based evaluation.

Domain	B1	EM	F1	B2	EM	F1	B3	EM	F1	B4	EM	F1	Model	EM	F1	B1 Rank
Islamism	Bert	0,72	0,77	GPT 3.5	0,72	0,77	GPT 4.1	0,51	0,62	Qwen2	0,31	0,41	KB Model	0.69	0.76	1
1973 oil crisis	Bert	0.68	0.74	GPT 3.5	0.48	0.60	GPT 4.1	0,61	0,66	Qwen2	0,45	0,53	KB Model	0.68	0.75	2
Kenya	Bert	0.66	0.76	GPT 3.5	0.43	0.56	GPT 4.1	0,44	0,54	Qwen2	0,28	0,43	KB Model	0.60	0.71	3
Harvard University	Bert	0.63	0.63	GPT 3.5	0.61	0.71	GPT 4.1	0,66	0,73	Qwen2	0,51	0,57	KB Model	0.71	0.77	4
Sky (United Kingdom)	Bert	0.62	0.66	GPT 3.5	0.50	0.62	GPT 4.1	0,65	0,74	Qwen2	0,36	0,43	KB Model	0.70	0.77	5
Civil disobedience	Bert	0.38	0.47	GPT 3.5	0.44	0.50	GPT 4.1	0,46	0,57	Qwen2	0,34	0,40	KB Model	0.47	0.52	44
European Union law	Bert	0.36	0.53	GPT 3.5	0.36	0.52	GPT 4.1	0,52	0,62	Qwen2	0,26	0,42	KB Model	0.55	0.64	45
Prime number	Bert	0.35	0.43	GPT 3.5	0.28	0.40	GPT 4.1	0,47	0,59	Qwen2	0,23	0,38	KB Model	0.37	0.49	46
Chloroplast	Bert	0.34	0.48	GPT 3.5	0.26	0.38	GPT 4.1	0,38	0,51	Qwen2	0,18	0,32	KB Model	0.52	0.59	47
Computational theory	Bert	0.34	0.51	GPT 3.5	0.26	0.38	GPT 4.1	0,42	0,54	Qwen2	0,14	0,25	KB Model	0.46	0.56	48

TABLE 19. KB model vs baselines with question type perspective, q type count ≥ 100.

Question Type	B1	EM	F1	B2	EM	F1	B3	EM	F1	B4	EM	F1	Model	EM	F1	B1 Rank
Ne zaman (When)	Bert	0,76	0,80	GPT 3.5	0.72	0.60	GPT 4.1	0.75	0.85	Qwen2	0,27	0,43	KB Model	0.81	0.75	1
Kaç (How much)	Bert	0.69	0.73	GPT 3.5	0.48	0.33	GPT 4.1	0.47	0.63	Qwen2	0,24	0,33	KB Model	0.73	0.65	2
Ne kadar (How much)	Bert	0.62	0.67	GPT 3.5	0.49	0.34	GPT 4.1	0.46	0.63	Qwen2	0,19	0,31	KB Model	0.57	0.44	3
Kim (Who)	Bert	0.57	0.63	GPT 3.5	0.72	0.62	GPT 4.1	0.73	0.79	Qwen2	0,57	0,65	KB Model	0.77	0.73	4
Nerede (Where)	Bert	0.45	0.52	GPT 3.5	0.55	0.43	GPT 4.1	0.51	0.63	Qwen2	0,19	0,34	KB Model	0.64	0.58	5
Hangi (Which)	Bert	0.51	0.59	GPT 3.5	0.65	0.54	GPT 4.1	0.65	0.73	Qwen2	0,46	0,56	KB Model	0.69	0.62	6
Ne (What)	Bert	0.42	0.52	GPT 3.5	0.48	0.36	GPT 4.1	0.50	0.61	Qwen2	0,28	0,39	KB Model	0.61	0.53	7

approaches. Numerous error analyses have encompassed microstudy areas that stand on their own.

Error Case 1: Generative models (GPTs) tend to return all types of entities that are most similar to the insight they get from the question.

- *Question:* Super Bowl 50 devre arası şovunu hangi grup manşete taşıdı? (Which band headlined the Super Bowl 50 halftime show?)
- *Correct Answer:* Coldplay
- *KB-Model Predict:* Coldplay
- *ChatGPT Predict:* Coldplay, Beyoncé ve Bruno Mars

Error Case 2: In machine translation, certain answers in the paragraph are displayed as numbers, while the response presents them in written form. This inconsistency leads to challenges in the assessment of the accuracy of a given answer.

- *Question:* Von Miller maç sırasında kaç top kaptı? (How many tackles did Von Miller get during the game?)
- *Correct Answer:* 5
- *KB-Model Predict:* beş (five)
- *ChatGPT Predict:* Beş (five)

Error Case 3: Models have difficulty with questions like "who were they" and "which were they". While Bert returns its first entity, OpenAI tends to return all the entities.

- *Question:* Super Bowl şampiyonluğunu savunan kimlerdi? (Who were the defending Super Bowl champions?)
- *Correct Answer:* New England Patriots
- *KB-Model Predict:* New England Patriots
- *ChatGPT Predict:* Denver Broncos ve New England Patriots

Error Case 4: Since Turkish is an agglutinative language, models have difficulty in using affix-based directions in questions. ("kimi" not equal "kim" in Turkish but "who" equals "kimi" and "kim" in English.)

- *Question 1:* Panthers NFC şampiyonluk maçında **kimi** yendi? (Who did the Panthers beat in the NFC championship game?)
- *Correct Answer:* Arizona Cardinals
- *KB-Model Predict:* New England Patriots
- *ChatGPT Predict:* Arizona Cardinals

- *Question 2:* AFC şampiyonası'nda Broncos'a **kim** yenildi? (Who lost to the Broncos in the AFC championship?)
- *Correct Answer:* New England Patriots
- *KB-Model Predict:* New England Patriots
- *ChatGPT Predict:* Arizona Cardinals

Error Case 5: In cases where ChatGPT cannot find the answer, it can give hallucinatory answers.

- Question :2015 NFL sezonunun en değerli oyuncusu kimdi? (Who was the most valuable player of the 2015 NFL season?)
- *Correct Answer:* Cam Newton
- *KB-Model Predict:* Cam Newton
- *ChatGPT Predict:* 2015, NFL, sezon, en değerli oyuncu, kim? (2015, NFL, season, Most Valuable Player, who?)

Error Case 6: Models have difficulty with questions involving inference, counting, or ordering.

- *Question:* Düşünülen üçüncü şehir neydi? (What was the third city considered?)
- *Correct Answer:* San Francisco
- *KB-Model Predict:* New Orleans
- *ChatGPT Predict:* New Orleans, Miami ve San Francisco (New Orleans, Miami and San Francisco)

Error Case 7: Models may encounter a dilemma when responding to inquiries that could be deemed accurate in either situation. This outcome adversely affects the Exact Match when a conclusive answer is sought.

- *Question:* WBT-FM hangi şehirde bulunur? (What city is WBT-FM located in?)
- *Correct Answer:* Chester, Güney Carolina (Chester, South Carolina)
- *KB-Model Predict:* Chester
- *ChatGPT Predict:* Chester, Güney Carolina (Chester, South Carolina)

Error Case 8: Models struggle with inquiries regarding the first or last. Owing to bias, the nearest semantic entity to the statement may be provided as an answer.

- *Question:* California en son ne zaman Super Bowl'a ev sahipliği yaptı? (When was the last time California hosted the Super Bowl?)
- *Correct Answer:* 2003
- *KB-Model Predict:* 2003
- *ChatGPT Predict:* 2003 ve 2014

Error Case 9: The model might struggle to identify the block when the proper nouns are mixed with the common nouns.

- *Question:* NFL'de hangi takım şimdiye kadarki en iyi başlangıcı yaptı? (Which team has had the best start ever in the NFL?)
- *Correct Answer:* Carolina Panthers
- *KB-Model Predict:* Carolina
- *ChatGPT Predict:* Panthers

Error Case 10: In generative models, formatting mistakes can happen because of insufficient reliance on the text.

- *Question:* Newton 2015'te kaç yard attı? (How many yards did Newton throw for in 2015?)
- *Correct Answer:* 3.837
- *KB-Model Predict:* 3.837
- *ChatGPT Predict:* 3837

Error Case 11: The generative model offered a superior fit for the outcomes related to the synonymy between the context and the response following the Turkish translation.

- *Question:* Broncos'un koçu kim? (Who is the head coach of the Broncos?), Head coach = Koç, Antrenör in Turkish but question contains "Koç", content contains "Antrenör".
- *Correct Answer:* Gary Kubiak
- *KB-Model Predict:* Wade Phillips
- *ChatGPT Predict:* John Fox ve Gary Kubiak

Error Case 12: Since Turkish is an agglutinative language, the responses generated by the models include suffixes. This influences the exact matching score.

- *Question:* Thomas Davis NFC Şampiyonluk Oyunu sırasında hangi vücut parçasını kırdı? (What body part did Thomas Davis break during the NFC Championship Game?)
- *Correct Answer:* kol (arm)
- *KB-Model Predict:* kolla (with arm)
- *ChatGPT Predict:* Kırık kolu (broken arm)

Error Case 13: It has been observed that a broad and comprehensive LLM model (ChatGPT) recognizes and interprets some technical terms better.

- Question :50. Super Bowl'un isimlendirilmesinde Roma rakamları kullanılsaydı hangisi kullanılırdı? (If Roman numerals were used in naming Super Bowl 50, which would they be?)
- *Correct Answer:* L
- *KB-Model Predict:* Super Bowl 50
- *ChatGPT Predict:* L

Error Case 14: In the answers sought in or outside the parentheses, the models focus more on returning all the data, in which case it affects the exact match.

- *Question:* Program kaç kg ay taşını geri getirdi? (How many kg of moon rocks did the program bring back?)
- *Correct Answer:* 382 kg
- *KB-Model Predict:* 842 pound (382 kg)
- *ChatGPT Predict:* 842 pound (382 kg)

Error Case 15: When asked when it ended, it started and how long it lasted in historical data given over a period of time, models generally provide the correct answer indirectly.

- *Question:* Watt, Newcomen'in motorundaki geliştirmelerinin gelişimini ne zaman bitirdi? (When did Watt complete her improvements on Newcomen's engine?)
- *Correct Answer:* 1775
- *KB-Model Predict:* 1763-1775
- *ChatGPT Predict:* 1763-1775

Error Case 16: It is one of the factors that affect the full accuracy in open-ended situations in the dataset.

- *Question:* Varşova'nın Eski Kent'i 1980'de hangi listeye yazıldı? (What list was Warsaw's Old Town included in 1980?)
- *Answer:* UNESCO'nun Dünya Mirası listesi (UNESCO's World Heritage List)

- *KB-Model Predict:* UNESCO
- *ChatGPT Predict:* UNESCO
- *Question:* Kimin tezi, bir sorunun çözümünün polinom zaman algoritmasına izin verdiğini varsayarak makul kaynaklarla çözülebilir olduğunu belirtir? (Whose thesis states that a problem is solvable with reasonable resources assuming that the solution allows a polynomial time algorithm?)
- *Answer:* Cobham'ın tezi (thesis of Cobham)
- *KB-Model Predict:* Cobham
- *ChatGPT Predict:* Cobham

Error Case 17: Models are generally difficult to answer, especially in formula-based questions that require deep scientific knowledge and interpretation skills.

- Question :Polinom süresi NP ile tamamlanmış bir sorun içinde kullanılabiliyorsa, P'nin neye eşit olduğu anlamına gelmez? (If polynomial time can be used in an NP-complete problem, does it not imply that P is equal to what?)
- *Answer:* NP
- *KB-Model Predict:* X sorunu C (X-C problem)
- *ChatGPT Predict:* P = NP

E. PERFORMANCE ANALYSIS

All models were tested on SQuAD test dataset with approximately 8,000 questions. The zero-shot prompt calling technique was used. Zero-shot calling is a way of interacting with a language model where you ask it to perform a task without providing any examples of how that task should be done. Zero-shot prompting offers several advantages: it is fast and easy to set up since it requires no task-specific examples, making it convenient when dealing with new or ad-hoc tasks. The performance and costs of the models during the testing process are shown in Table 20.

In terms of performance, BERT-based models (Baseline1 and KBModel) are highly efficient for response time. At the same time, these models are also cost-effective because they can be hosted self-hosted on an end-user computer. ChatGPT-3.5 is more expensive and slower than the BERT-based models in terms of performance and cost, but compared to ChatGPT-4.5, it is both a faster model and a lower-cost model. Qwen2-7B, as an open source alternative, while being faster and substantially more cost-effective when self-hosted, makes it attractive for cloud deployment and scaling.

TABLE 20. Performance of models 8K question call for test dataset.

Model	Total Response Time	Cost (API or self-host)
GPT-3.5	28 Hours	25 (Open AI)$
GPT-4.1	63 Hours	200 (Open AI) $
Qwen2-7B-Instruct	14 Hours	1 $ (Google colab)
BERT	3 Hours	self-host
KBModel	3 Hours	self-host

VI. ALTERNATIVE DATASET RESULTS

As an alternative to KBModel with the mSQuAD dataset, we compared it with the existing TQuADv1 dataset. This dataset was used to observe the performance of KBModel with different models run on TQuADv1 [18]. The TQuADv1 dataset has already been prepared as a reading-comprehension dataset for Turkish. However, this dataset is only about Turkish & Islamic Science History and has a very limited number of questions. The dataset contained only 892 test questions. The KBModel was tested using this dataset. Compared with the outputs of [18], which worked on this dataset, KBModel achieved more successful results.

The model achieved a high success rate of 0.60 as Exact Match and 0.77 as F1 score. Owing to the limited number of topics and questions, a drill-down analysis such as mSQUAD was not performed. Reference [18] study performed different fine tuning operations on BERTurk-base, T5-small, T5-base, T5-large with TQuADv1 dataset. We compare the results of evaluating the KBModel outputs with the models in Table 21.

TABLE 21. Performance of KBModel on TQuADv1 dataset and comparison with model success of [18] study.

Model	EM	F1
BERTurk-base	0.45	0.62
MT-Both T5-small	0.48	0.63
MT-Both T5-base	0.55	0.72
MT-Both T5-large	0.59	0.74
KBModel	0.60	0.77

VII. CONCLUSION

The KB Model significantly outperforms both Baseline1 (BERT) and Baseline2 (ChatGPT 3.5) across various token ranges, question types, and domains, demonstrating enhancements of 6-18% in Exact Match (EM) and 6-17% in F1 scores, particularly in answer-oriented tasks and achieving notable gains in both the top five and bottom five domains for Baseline1. Compared to Baseline3 (ChatGpt 4.1), the KBModel F1 score (+%3) and the exact match (+%1) were generally superior. Although the KB Model produced more successful results, Baseline3 sometimes produced better F1 scores in limited cases. Baseline4 (Qwen2) model produced less accurate results than all baselines.

Transforming the data into a structured format and enriching the question information positively increased the success of the model. It provided a more balanced and successful model in cases such as domain-by-domain, long token length, and question type fluctuating in the baselines. Creating more comprehensive and accurate knowledge-based models will enable this success to be achieved at a higher level. The infrastructure used to create the knowledge base (LangChain + OpenAI + ChatGPT) was used as the service and framework. This infrastructure is critical for knowledge mapping and structuring. When the infrastructure has multilingual support, it can be installed in other languages. However, infrastructure currently runs as SAAS; therefore, providing this infrastructure as an open source is an important area of work for the future.

In error analyses, there are special cases where both span classification and generative AI are inadequate. There are areas of development for all the models that require more intelligence, such as inference, ordering, selection, and counting. In addition, varying accuracies have been detected, such as language structure, spelling, translation output format, and synonymy. However, each issue requires detailed research.

A reference study on the use of the translation method was conducted in languages with few sample datasets. The problems and solutions brought about by the translation method were shared. Simultaneously, the accuracy of the models was observed in deep learning models with the structural format and enrichment of the data obtained. It has been shown that effective results can be obtained on the main basis with data structuring of LLM and Deep Learning models in a black box structure. It has shown the importance of structuring data in question and answer systems.

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